

Ultimate Nutrition

Super Vitamin B-Complex

The benefits of vitamins are not news, but we need them as surely as we need air to breathe and water to drink. Make no mistake; a good supplement regimen starts with adequate vitamin intake. B Vitamins are essential for maintaining healthy nerves, skin, hair, eyes, liver and mouth and also good for muscle tone in the gastrointestinal tract. They give us energy and are necessary to metabolize carbohydrates, fats and proteins. Because B vitamins are water soluble, our bodies naturally flush them away everyday. Physical and emotional stress, dieting and our typical "on-the-go" lifestyles also increase their loss. That is why it is so important to replenish them daily.

The B-complex vitamins are a group of similarly structured compounds, which is why we refer to them as a "complex". This complex of chemicals consists of B1 (thiamine), B-2 (riboflavin), B-3 (niacin), B-5 (pantothenic acid), B-6 (pyridoxine), B-7 (biotin), B-12 (cobalamin), and folic acid (folate or folacin). PABA, inositol and choline are often included in this group. The B-complex of vitamins are used in the proper formation of every cell in your body - particularly nerve cells. This is why it is so important for pregnant women to get their B's (especially folic acid).

The most common nutritional deficiency in the world is folic acid deficiency, which is characterized by low mood, and can lead to neural tube defects in the developing, unborn child. Deficiency in either vitamin B-12 or B-6 can also bring about low mood, and often mask folic acid deficiency. B vitamins are not stored well in the body and are depleted rapidly in dieters, and caffeine and alcohol users. The B-complex is used throughout the body in methylation reactions and is particularly effective against low mood, moodiness, and muscular weakness.

B vitamins help the body to maintain:

- health of the skin, bones, hair and muscle
- health of the mucosal membranes, particularly around the mouth
- intestinal health and bowel function
- brain cell function and health
- relief from skin problems, including dry and itchy skin and rashes

Because they are methyl donors, the B-complex helps to reduce the build-up of homocysteine, making it an excellent heart-health supplement. It is well established that a person's measure of blood plasma homocysteine acts as a biochemical marker inversely indicating folate levels. When a person's blood plasma folate levels are high, their homocysteine levels are low.

B-6 (pyridoxine). One of the water-soluble vitamins, Vitamin B-6 is needed for a wide array of bodily functions, particularly in the handling of amino acid reactions. Here are some of the more interesting functions B-6 performs in the body: metabolizes cysteine to pyruvic acid, works with phosphorylase to convert glycogen to glucose, works to synthesize fatty acids and metabolize cholesterol, is used to produce a variety of neurotransmitters including serotonin.

B-6 supports the immune system. While the B vitamins are found in foods, some are lost in cooking and excess of one B can result in a depletion of another. This is one vitamin we suggest taking as a complete supplement.

References

1. Shimakawa T, Nieto FJ, Malinow MR, Chambless LE, Schreiner PJ, Szklo M. Vitamin intake: a possible determinant of plasma homocyst(e)ine among middle-aged adults. *Ann Epidemiol* 1997;7:285-93.
2. Boushey CJ, Beresford SA, Omenn GS, Motulsky AG. A quantitative assessment of plasma homocysteine as a risk factor for vascular disease: probable benefits of increasing folic acid intakes. *JAMA* 1995;274:1049-57.
3. Nygard O, Nordrehaug JE, Refsum H, Ueland PM, Farstad M, Vollset SE. Plasma homocysteine levels and mortality in patients with coronary artery disease. *N Engl J Med* 1997;337:230-6.
4. Lauritzen, C., H.D. Reuter, R. Repges, K.-J. Bohnert, U. Schmidt. 1997. Treatment of premenstrual tension syndrome with Vitex agnus-castus. Controlled, double-blind study versus pyridoxine. *Phytomedicine*, Vol. 4, No. 3, 183-189.
5. Bernstein, A, Dinesen J: Brief communication: Effect of pharmacologic doses of vitamin B6 on carpal tunnel syndrome, electroencephalographic results, and pain. *J Am Col N* 1993;12:73-76.
6. American Journal of Obstetric Gynecology 1995; 173:881 84.